KUTAISI INTERNATIONAL UNIVERSITY

Scientific Computing

Ramaz Botchorishvili

Kutaisi International University

2.4: convergence of explicit Euler

Outline

Lecture 2.4: convergence of explicit Euler

- Setting: Lipschitz f, bounded u''
- Theorem: global error bound and order 1
- Proof in six steps
- Q & A

Series: 2.1 Euler schemes. 2.2 errors. 2.3 geometry. 2.4 explicit convergence. 2.5 Grönwall. 2.6 implicit convergence.

Problem: setting for the convergence analysis

Problem 2.1 (Assumptions for convergence: Lipschitz f, $u \in C^2$ with $\|u''\| \leq M$, grid $t_n = \alpha + nh$)

$$u'(t) = f(t, u(t)), \quad t \in [\alpha, \beta],$$
$$u(\alpha) = y_0.$$

Problem: setting for the convergence analysis

Problem 2.1 (Assumptions for convergence: Lipschitz f, $u \in C^2$ with $\|u''\| \leq M$, grid $t_n = \alpha + nh$)

$$u'(t) = f(t, u(t)), \quad t \in [\alpha, \beta],$$
$$u(\alpha) = y_0.$$

- $f \in C([\alpha, \beta] \times \mathbb{R}^n, \mathbb{R}^n)$ is Lipschitz in the second argument: $\|f(t, x) - f(t, z)\| \leq L\|x - z\|$.

Problem: setting for the convergence analysis

Problem 2.1 (Assumptions for convergence: Lipschitz f, $u \in C^2$ with $\|u''\| \le M$, grid $t_n = \alpha + nh$)

$$u'(t) = f(t, u(t)), \quad t \in [\alpha, \beta],$$
$$u(\alpha) = y_0.$$

- $f \in C([\alpha, \beta] \times \mathbb{R}^n, \mathbb{R}^n)$ is Lipschitz in the second argument: $\|f(t, x) - f(t, z)\| \le L\|x - z\|$.
- The unique solution satisfies $u \in C^2([\alpha, \beta])$ and $\|u''(t)\| \le M$.

Problem: setting for the convergence analysis

Problem 2.1 (Assumptions for convergence: Lipschitz f, $u \in C^2$ with $\|u''\| \le M$, grid $t_n = \alpha + nh$)

$$u'(t) = f(t, u(t)), \quad t \in [\alpha, \beta],$$
$$u(\alpha) = y_0.$$

- $f \in C([\alpha, \beta] \times \mathbb{R}^n, \mathbb{R}^n)$ is Lipschitz in the second argument: $\|f(t, x) - f(t, z)\| \le L\|x - z\|$.
- The unique solution satisfies $u \in C^2([\alpha, \beta])$ and $\|u''(t)\| \le M$.
- Step size h, grid $t_n = \alpha + nh$, error $e_n = y_n - u(t_n)$.

Theorem: convergence of explicit Euler

Theorem 2.2 (Convergence of explicit Euler with error bound $\frac{M}{2L}\left(e^{L(t_n-\alpha)}-1\right)h$)

In the setting above let $y_{n+1} = y_n + h\,f(t_n, y_n)$, $t_n = \alpha + nh$. Then

$$\|e_n\| \leq \frac{M}{2L}\left(e^{L(t_n-\alpha)}-1\right)h.$$

Theorem: convergence of explicit Euler

Theorem 2.2 (Convergence of explicit Euler with error bound $\frac{M}{2L}\left(e^{L(t_n-\alpha)}-1\right)h$)

In the setting above let $y_{n+1} = y_n + h\,f(t_n, y_n)$, $t_n = \alpha + nh$. Then

$$\|e_n\| \le \frac{M}{2L}\left(e^{L(t_n-\alpha)}-1\right)h.$$

Consequently $\|e_n\| = O(h)$: the method converges with order 1.

Theorem: convergence of explicit Euler

Theorem 2.2 (Convergence of explicit Euler with error bound $\frac{M}{2L}\left(e^{L(t_n-\alpha)}-1\right)h$)

In the setting above let $y_{n+1} = y_n + h\,f(t_n, y_n)$, $t_n = \alpha + nh$. Then

$$\|e_n\| \le \frac{M}{2L}\left(e^{L(t_n-\alpha)}-1\right)h.$$

Consequently $\|e_n\| = O(h)$: the method converges with order 1.

Plan of the proof

LTE $\to$ error recurrence $\to$ unfold $\to$ bound the LTE $\to$ geometric sum $\to$ result.

Proof of the convergence theorem, step 1: LTE

Proof of explicit Euler convergence, step 1: local error $\tau_{n+1} = O(h^2)$.

- Taylor expansion of the exact solution, $\xi_{n+1} \in (t_n, t_{n+1})$:

$$u(t_{n+1}) = u(t_n) + h\,u'(t_n) + \frac{h^2}{2}u''(\xi_{n+1}).$$

(for vector u the remainder is taken componentwise or in integral form.)

Proof of the convergence theorem, step 1: LTE

Proof of explicit Euler convergence, step 1: local error $\tau_{n+1} = O(h^2)$.

- Taylor expansion of the exact solution, $\xi_{n+1} \in (t_n, t_{n+1})$:

$$u(t_{n+1}) = u(t_n) + h\,u'(t_n) + \frac{h^2}{2}u''(\xi_{n+1}).$$

  (for vector u the remainder is taken componentwise or in integral form.)

- With $u'(t_n) = f(t_n, u(t_n))$ and $\tau_{n+1} = \frac{h^2}{2}u''(\xi_{n+1})$:

$$u(t_{n+1}) = u(t_n) + h\,f(t_n, u(t_n)) + \tau_{n+1}.$$

Proof of the convergence theorem, step 1: LTE

Proof of explicit Euler convergence, step 1: local error $\tau_{n+1} = O(h^2)$.

- Taylor expansion of the exact solution, $\xi_{n+1} \in (t_n, t_{n+1})$:

$$u(t_{n+1}) = u(t_n) + h\,u'(t_n) + \frac{h^2}{2}u''(\xi_{n+1}).$$

  (for vector u the remainder is taken componentwise or in integral form.)

- With $u'(t_n) = f(t_n, u(t_n))$ and $\tau_{n+1} = \frac{h^2}{2}u''(\xi_{n+1})$:

$$u(t_{n+1}) = u(t_n) + h\,f(t_n, u(t_n)) + \tau_{n+1}.$$

- Hence the LTE is $\tau_{n+1} = O(h^2)$.

□

Proof of the convergence theorem, step 2: error recurrence

Proof of explicit Euler convergence, step 2: recurrence $\|e_{n+1}\| \le (1+hL)\|e_n\| + \|\tau_{n+1}\|$.

- Subtract the Euler step from the Taylor identity:

$$\underbrace{u(t_{n+1}) - y_{n+1}}_{-e_{n+1}} = \underbrace{u(t_n) - y_n}_{-e_n} + h \underbrace{\big(f(t_n, u(t_n)) - f(t_n, y_n)\big)}_{\text{Lipschitz term}} + \tau_{n+1}.$$

Proof of the convergence theorem, step 2: error recurrence

Proof of explicit Euler convergence, step 2: recurrence $\|e_{n+1}\| \le (1+hL)\|e_n\| + \|\tau_{n+1}\|$.

- Subtract the Euler step from the Taylor identity:

$$\underbrace{u(t_{n+1}) - y_{n+1}}_{-e_{n+1}} = \underbrace{u(t_n) - y_n}_{-e_n} + h \underbrace{\big(f(t_n, u(t_n)) - f(t_n, y_n)\big)}_{\text{Lipschitz term}} + \tau_{n+1}.$$

- Take norms and use $\|f(t_n, u(t_n)) - f(t_n, y_n)\| \le L\|e_n\|$:

$$\|e_{n+1}\| \le (1 + hL)\,\|e_n\| + \|\tau_{n+1}\|.$$

□

Proof of the convergence theorem, step 3: unfolding

Proof of explicit Euler convergence, step 3: unfolding the recurrence.

- Start value: $e_0 = y_0 - u(\alpha) = 0$.

Proof of the convergence theorem, step 3: unfolding

Proof of explicit Euler convergence, step 3: unfolding the recurrence.

- Start value: $e_0 = y_0 - u(\alpha) = 0$.
- Apply the recurrence repeatedly:

$$\|e_1\| \le \|\tau_1\|,$$
$$\|e_2\| \le (1 + hL)\|\tau_1\| + \|\tau_2\|,$$
$$\|e_n\| \le (1 + hL)^n \|e_0\| + \sum_{k=0}^{n-1} (1 + hL)^k \|\tau_{n-k}\| = \sum_{k=0}^{n-1} (1 + hL)^k \|\tau_{n-k}\|.$$

Proof of the convergence theorem, step 3: unfolding

Proof of explicit Euler convergence, step 3: unfolding the recurrence.

- Start value: $e_0 = y_0 - u(\alpha) = 0$.
- Apply the recurrence repeatedly:

$$\|e_1\| \le \|\tau_1\|,$$
$$\|e_2\| \le (1+hL)\|\tau_1\| + \|\tau_2\|,$$
$$\|e_n\| \le (1+hL)^n\|e_0\| + \sum_{k=0}^{n-1}(1+hL)^k\|\tau_{n-k}\| = \sum_{k=0}^{n-1}(1+hL)^k\|\tau_{n-k}\|.$$

- $1 + x \le e^x$ gives $(1+hL)^k \le e^{hLk}$, hence

$$\|e_n\| \le \sum_{k=0}^{n-1} e^{Lhk}\,\|\tau_{n-k}\|.$$



Proof of the convergence theorem, step 4: bound on the LTE

Proof of explicit Euler convergence, step 4: bounding each local error by $Mh^2/2$.

- From $\|u''\| \leq M$: $\|\tau_k\| \leq \dfrac{Mh^2}{2}$ for every k.

Proof of the convergence theorem, step 4: bound on the LTE

Proof of explicit Euler convergence, step 4: bounding each local error by $Mh^2/2$.

- From $\|u''\| \le M$: $\|\tau_k\| \le \dfrac{Mh^2}{2}$ for every k.
- Insert into the unfolded bound:

$$\|e_n\| \le \frac{Mh^2}{2} \sum_{k=0}^{n-1} e^{Lhk}.$$

Proof of the convergence theorem, step 4: bound on the LTE

Proof of explicit Euler convergence, step 4: bounding each local error by $Mh^2/2$.

- From $\|u''\| \le M$: $\|\tau_k\| \le \dfrac{Mh^2}{2}$ for every k.
- Insert into the unfolded bound:

$$\|e_n\| \le \frac{Mh^2}{2} \sum_{k=0}^{n-1} e^{Lhk}.$$

- The error is a weighted sum of all earlier local errors; the weights e^{Lhk} grow with the age k of the error.



Proof of the convergence theorem, step 5: geometric sum

Proof of explicit Euler convergence, step 5: summing the geometric series.

- Geometric series:

$$\sum_{k=0}^{n-1} e^{Lhk} = \frac{e^{Lhn} - 1}{e^{Lh} - 1}, \qquad e^{Lhn} = e^{L(t_n - \alpha)}.$$

Proof of the convergence theorem, step 5: geometric sum

Proof of explicit Euler convergence, step 5: summing the geometric series.

- Geometric series:

$$\sum_{k=0}^{n-1} e^{Lhk} = \frac{e^{Lhn}-1}{e^{Lh}-1}, \qquad e^{Lhn} = e^{L(t_n-\alpha)}.$$

- Taylor: $e^x - 1 \geq x$ for $x \geq 0$, so $e^{Lh} - 1 \geq Lh$ and $\dfrac{1}{e^{Lh}-1} \leq \dfrac{1}{Lh}$.

Proof of the convergence theorem, step 5: geometric sum

Proof of explicit Euler convergence, step 5: summing the geometric series.

- Geometric series:

$$\sum_{k=0}^{n-1} e^{Lhk} = \frac{e^{Lhn} - 1}{e^{Lh} - 1}, \qquad e^{Lhn} = e^{L(t_n - \alpha)}.$$

- Taylor: $e^x - 1 \geq x$ for $x \geq 0$, so $e^{Lh} - 1 \geq Lh$ and $\dfrac{1}{e^{Lh} - 1} \leq \dfrac{1}{Lh}$.
- Substitute:

$$\|e_n\| \leq \frac{Mh^2}{2} \cdot \frac{e^{L(t_n - \alpha)} - 1}{Lh}.$$

□

Proof of the convergence theorem, step 6: conclusion

Proof of explicit Euler convergence, step 6: final bound and first order.

- Simplify:

$$\|e_n\| \leq \frac{M}{2L}\left(e^{L(t_n-\alpha)}-1\right)h.$$

Proof of the convergence theorem, step 6: conclusion

Proof of explicit Euler convergence, step 6: final bound and first order.

- Simplify:

$$\|e_n\| \leq \frac{M}{2L}\Big(e^{L(t_n-\alpha)} - 1\Big)h.$$

- M and L do not depend on h and $t_n \leq \beta$, so

$$\|e_n\| \leq C\,h, \qquad C = \frac{M}{2L}\Big(e^{L(\beta-\alpha)} - 1\Big).$$

Proof of the convergence theorem, step 6: conclusion

Proof of explicit Euler convergence, step 6: final bound and first order.

- Simplify:

$$\|e_n\| \leq \frac{M}{2L}\left(e^{L(t_n-\alpha)}-1\right)h.$$

- M and L do not depend on h and $t_n \leq \beta$, so

$$\|e_n\| \leq C\,h, \qquad C=\frac{M}{2L}\left(e^{L(\beta-\alpha)}-1\right).$$

- The global error is proportional to h: explicit Euler is first-order convergent.

□

References

- **Euler scheme error.** N. Czówska, P. M. Morkisz, P. Przybyłowicz, *Euler scheme for approximation of solution of nonlinear ODEs under inexact information*, `arXiv:2209.07482` (ref.).

Q & A